

Supplementary Information for Benchmark-Driven Multivariate Time Series Classification: The Role of the UEA MTSC Archive

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1 Introduction

This supplementary document provides supporting reference material for the survey. Table 2 to Table 7 list the abbreviated algorithm names used throughout the paper, together with expanded or descriptive names and their references. Where a model name is stylised rather than a strict acronym, we give a brief descriptive form instead. The survey also uses a number of common acronyms for tasks, data types, model families, and baseline methods. These are collected in Table 1, together with their expanded forms.

2 Common Acronyms Used in the Survey

Table 1: Common acronyms and abbreviated method names used in the survey. This table focuses on generic abbreviations and widely used legacy method acronyms that appear in the text. Model names from individual papers are left in their published form when they are better treated as proper names than as standard abbreviations.

Acronym	Expanded name
MTSC	Multivariate time series classification
TSC	Time series classification
MTS	Multivariate time series
UEA	University of East Anglia
UCR	University of California, Riverside
CSV	Comma-separated values
ECG	Electrocardiography / electrocardiogram
EEG	Electroencephalography / electroencephalogram
MEG	Magnetoencephalography / magnetoencephalogram
IMU	Inertial measurement unit
CNN	Convolutional neural network

Acronym	Expanded name
TCN	Temporal convolutional network
FCN	Fully convolutional network
RNN	Recurrent neural network
GRU	Gated recurrent unit
SVM	Support vector machine
GNN	Graph neural network
GCN	Graph convolutional network
GAT	Graph attention network
LLM	Large language model
DCT	Discrete cosine transform
ED	Euclidean distance
DTW	Dynamic time warping
DTW-D	Dependent dynamic time warping
DTW-I	Independent dynamic time warping
SAX	Symbolic Aggregate approXimation
TSF	Time Series Forest
CIF	Canonical Interval Forest
DrCIF	Diverse Representation Canonical Interval Forest
HIVE-COTE	Hierarchical Vote Collective of Transformation-Based Ensembles
HC1	HIVE-COTE 1.0
ROCKET	RandOm Convolutional KErnel Transform
STC	Shapelet Transform Classifier
TDE	Temporal Dictionary Ensemble
CAWPE	Cross-validation Accuracy Weighted Probabilistic Ensemble
cBOSS / CBOSS	Contractable Bag-of-SFA-Symbols
WEASEL	Word ExtrAction for time SEries cLassification
MUSE	Multivariate Unsupervised Symbols and dErivatives

Table 2: Machine-learning models: short names and expanded or descriptive names.

Short name	Expanded or descriptive name	Reference
CIF	Canonical Interval Forest	(Middlehurst et al., 2020)
MSAX	Multivariate Symbolic Aggregate approXimation	(Anacleto et al., 2020)
MI-FSS	Mutual-information feature selection strategy	(Ircio et al., 2020)
MSTS	MSTS feature-selection method (name as given by the authors)	(Kathirgamanathan and Cunningham, 2020)
CMFMTS	Complexity-measure feature model for multivariate time series	(Baldán and Benítez, 2021)
F4	F4 feature-based classifier (name as given by the authors)	(López-Oriona and Vilar, 2021)
HC2	HIVE-COTE 2.0	(Middlehurst et al., 2021)
MTSPF	Multivariate Time Series Proximity Forest	(Zhang et al., 2021a)
RT	Rule Transform	(Bahri et al., 2022)
PETSC	PETSC pattern-mining classifier (name as given by the authors)	(Feremans et al., 2022)
EP / IEP	Ensemble-prediction framework / improved ensemble-prediction framework	(Campagner et al., 2024)
FS-MVC	Feature-selection and multiview classification method	(He et al., 2024)
Z-Time	Z-Time symbolic temporal abstraction model	(Lee et al., 2024)
EE-TSS	Engineered-feature time-series signature method	(Patharkar et al., 2024)
BORF	Bag-of-Receptive-Fields	(Spinnato et al., 2024)
BOP	Bag of Patterns	(Wang et al., 2024d)
FedDict	Federated dictionary-based classifier	(Liang et al., 2025a)
GSM	Signature and logsignature model	(Panwar, 2025)
TimeCIEL	Time-series contextual interactive ensemble learning	(Levy et al., 2026)

Table 3: Convolutional models: short names and expanded or descriptive names.

Short name	Expanded or descriptive name	Reference
CA-SFCN	Cross-attention spatial fully convolutional network	Hao and Cao (2020)
CT_CAM	Channel-temporal attention model	Cheng et al. (2020)
TapNet	Attentional prototypical network for multivariate time series	Zhang et al. (2020)
Disjoint-CNN	Disjoint convolutional neural network	Foumani et al. (2021)
GM-FCN	GM-FCN convolutional model (name as given by the authors)	Tai et al. (2021)
XCM	Explainable Convolutional Neural Network for Multivariate Time Series Classification	Fauvel et al. (2021)
AC-LSTM	AC-LSTM hybrid convolutional recurrent model	Zhang et al. (2022)
DA-Net	Dual-attention network	Chen et al. (2022)
dCAM	Dimension-aware class activation map model	Boniol et al. (2022)
DSN	Dynamic sparse network	Xiao et al. (2022)
Multichannel CNN	Multichannel convolutional neural network	Oh (2022)
TCRAN	Temporal-correction residual attention network	Zhu et al. (2022)
Adacket	Adacket neural architecture search model	Zhang et al. (2023)
Conv-GLU	Convolutional gated linear unit network	Liu et al. (2023)
POPNASv3	POPNAS version 3 neural architecture search model	Falanti et al. (2023)
AdaFSNet	Adaptive feature-selection network	Wang et al. (2024b)
TPS	Temporal plug-in attention module	Abbasi and Saeedi (2024)
DCA	DCT-based channel attention model	Haboub et al. (2025)
DM-FCN	Dilated convolution fully convolutional network variant	Zhang et al. (2025)
FQN	Quanvolutional network	Orka et al. (2025)
FreConvNet	Frequency-aware convolutional network	Tang et al. (2025b)
HDTCNet	Hybrid-dimension temporal convolution network	Gu (2025a)
LITEMVTime	LITE multivariate time-series classifier	Ismail-Fawaz et al. (2025)
RE-ModernTCN	Re-evaluated Modern Temporal Convolutional Network	Akacik and Hoogendoorn (2025)

Short name	Expanded or descriptive name	Reference
SCNet	Spectral convolution network	Wu et al. (2025b)
TEAM	TEAM convolutional model (name as given by the authors)	Qiu et al. (2025b)

Table 4: Graph-based models: short names and expanded or descriptive names.

Short name	Expanded or descriptive name	Reference
MTPool	Multivariate time-series pooling model	Duan et al. (2022)
SimTSC	Similarity-based Time Series Classification	Zha et al. (2022)
CATodyNet	CATodyNet dynamic temporal graph model	Gui et al. (2024)
HyperTime	HyperTime hypergraph temporal model	Younis and Ahmadi (2024)
MagNet	MagNet graph model	Hong et al. (2024)
TF-Net	Time-frequency network	Lei et al. (2024)
RTNet	Relational temporal network	Gu (2024)
CTSA	CTSA heterogeneous graph model	Du and Chen (2025)
DET-DyAGCN	Dynamic graph attention convolution model	Gui (2025)
DVG-CL	Dual-view graph contrastive learning	Xiao et al. (2025b)
GSTT	GSTT graph temporal model	Zhang et al. (2026)
KDP-MHL	Key-point-aware multi-hypergraph learning	Ma et al. (2025a)
MH-TFFN	Mamba-hypergraph time-frequency fusion network	Jiang et al. (2025)
MuPaST	Multi-period spatio-temporal model	Li et al. (2025)
RP-GCN	Recurrence-plot graph convolutional network	Kang et al. (2025)
STOP	STOP graph model	Liu et al. (2025)
TV-Net	Temporal-variable network	Yue et al. (2025)

Table 5: Transformer-based models: short names and expanded or descriptive names.

Short name	Expanded or descriptive name	Reference
FlowFormer	FlowFormer transformer	Wu et al. (2022)
FourierFormer	FourierFormer transformer	Nguyen et al. (2022b)
GFISHformer	GFISHformer transformer	Nguyen et al. (2022a)
FormerTime	FormerTime transformer	Cheng et al. (2023)
MCTNet	Multichannel transformer network	Wu et al. (2023b)

Short name	Expanded or descriptive name	Reference
MF-Net	Multi-feature network	Du et al. (2023)
Primal-Attention	Primal-attention transformer	Chen et al. (2023)
AutoLDT	Lightweight decoupled transformer with AutoML	Wang et al. (2024c)
ConvTran	Convolutional Transformer	Foumani et al. (2024b)
CTNet	Crucial-timestamp network	Zhang et al. (2024)
DCSformer	Dynamic-chunking transformer	Wang et al. (2024a)
DTCM	DTCM transformer with capsule mutual distillation	Xiao et al. (2024b)
MEDT	Multiview encoder-decoder transformer	Li et al. (2024)
MSTformer	Multiscale spatial-temporal transformer	Wu et al. (2024)
STIFormer	Spatial-temporal interaction transformer	Zhang and Li (2024)
ShapeFormer	Shapelet transformer	Le et al. (2024)
CAB	Correlated-attention block transformer	Nguyen et al. (2025)
CSTFormer	Cross-scale temporal transformer	Qiu et al. (2025a)
DTWformer	Dynamic Time Warping transformer	Yu et al. (2025)
Hformer	Hierarchical transformer	Tang et al. (2025a)
KATN	Kernel-attention transformer network	Xiao et al. (2025a)
MD-Former	MD-Former transformer	Du et al. (2025)
PBC-CSA	Prototype-based attention transformer	Gu (2025b)
PatchMTSC	Patch-based transformer for multivariate time series classification	Wei et al. (2025)
SGN	SGN transformer model	Ying et al. (2025)
Sliding-window Transformer + DTW similarity	Sliding-window transformer with DTW similarity	Wu et al. (2025a)
TFTN	Time-frequency transformer network	Huang et al. (2026)

Table 6: Foundation-style models: short names and expanded or descriptive names.

Short name	Expanded or descriptive name	Reference
TST	Time Series Transformer	Zerveas et al. (2021)
TARNet	Task-aware representation network	Chowdhury et al. (2022)
TS2Vec	Time Series to Vector	Yue et al. (2022)
FEAT	Feature-aware self-supervised representation learner	Kim et al. (2023)
TimesNet	TimesNet general-purpose backbone	Wu et al. (2023a)
HSC-MAE	Hierarchical sparse-convolutional masked autoencoder	Yu et al. (2024)
LPTM	Large pre-trained time-series model	Kamarthi and Prakash (2024)
LSTST	LSTST language-model-based time-series classifier	Jang et al. (2024)
ModernTCN	Modern temporal convolutional network	Luo and Wang (2024)
TimesURL	Time-series universal representation learning	Liu and Chen (2024)
Series2Vec	Series to Vector	Foumani et al. (2024a)
CBD	CBD foundation-style backbone	Han et al. (2025)
CauLLM	Causality-aware large language model for time series	Wu et al. (2025c)
LETS-C	LETS-C language-enhanced time-series classifier	Kaur et al. (2025)
LLMFew	LLMFew few-shot large-language-model classifier	Chen et al. (2025b)
PETS	Pattern-assisted multi-task time-series model	Ma et al. (2025b)
S2TS-LLM	Sequence-to-time-series large language model	Qin et al. (2025)
TableTime	TableTime tabular in-context time-series model	Wang et al. (2025)
TFAS	Time-series foundation architecture search	Trirat and Lee (2025)
TSFM Adapters	Time-series foundation model adapters	Ilbert et al. (2025)
HiTime	HiTime multimodal time-series model	Tao et al. (2026)

Table 7: Other deep models: short names and expanded or descriptive names.

Short name	Expanded or descriptive name	Reference
STSP-SNN	Spiking neural network for spatio-temporal spike patterns	Fang et al. (2020)
ShapeNet	Differentiable shapelet network	Li et al. (2021)

Short name	Expanded or descriptive name	Reference
GMSM _{red}	Neural multivariate shapelet model	Medico et al. (2021)
RTFN	RTFN recurrent-temporal fusion model	Xiao et al. (2021)
mGRN	Multivariate gated recurrent network	Zhang et al. (2021b)
SMATE	SMATE recurrent spatial-interaction model	Zuo et al. (2021)
WHEN	Wavelet and DTW-inspired hybrid model	Wang et al. (2023)
GP-HLS	GP-HLS representation-learning model	Ye and Ma (2023a)
LBP4MTS	LBP4MTS representation-learning model	Ye and Ma (2023b)
DSC	Federated semantic communication model for MTSC	Zhao et al. (2023)
Chimera	Chimera state-space model	Behrouz et al. (2024)
AGDM	Adaptive-granularity dimension-decoupling model	Ding et al. (2024)
FDNet	Feature-difference network	Gao et al. (2024)
MTSFNet	Multivariate temporal-spectral fusion network	Huang et al. (2024)
DECM	Deep Efficient Continuous Manifold Learning	Jeong et al. (2024)
DSDCLNet	Dual-stream contrastive learning network	Liu et al. (2024)
RankSCL	Rank-supervised contrastive learning	Ren et al. (2024)
VQShape	Vector-quantised shapelet model	Wen et al. (2024)
DKN	Densely knowledge-aware network	Xiao et al. (2024a)
TSec	Efficient time-series classifier	Yao et al. (2024)
TSCMamba	Mamba-based time-series classifier	Ahamed and Cheng (2025)
LETO	LETO hybrid MTSC model	Behrouz et al. (2025)
SFC	SFC frequency-aware hybrid model	Chen et al. (2025a)
DAPNet	Dynamic parsing network	Gao (2025)
TCFNet	Time-channel fusion network	Liang et al. (2025b)
RFACL	Frequency-aware contrastive learning model	Lin et al. (2025)
MPTSNet	MPTSNet hybrid MTSC model	Mu et al. (2025)
GrowingNN	Adaptive growing neural network	Swiderski et al. (2025)
FreRA	Frequency-aware representation and augmentation model	Tian et al. (2025)
NI-MTSC	Noise-/uncertainty-informed MTSC model	Yang et al. (2025)
RTF	RTF time-frequency contrastive model	Yuan et al. (2026)

Short name	Expanded or descriptive name	Reference
MatrixMixer	MatrixMixer selective state-space model	Azizi and BabaAli (2026)
WaveState	WaveState state-space model	Li et al. (2026)

3 Machine Learning Algorithms

Table 8: Accuracy of five machine-learning classifiers on 26 equal length UEA datasets used to draw Figure ?? . The highest accuracy in each row is shown in bold.

Dataset	HC2	CMFM	RT	BORF	CIF
ArticularyWordRecognition	0.993	0.988	0.993	0.993	0.983
AtrialFibrillation	0.267	0.191	0.200	0.267	0.333
BasicMotions	1.000	0.982	1.000	0.975	1.000
Cricket	1.000	0.977	1.000	1.000	0.986
DuckDuckGeese	0.560	0.510	0.180	0.440	0.440
ERing	0.989	0.931	0.8185	0.952	0.981
EigenWorms	0.947	0.895	0.5878	0.870	0.916
Epilepsy	1.000	0.999	0.9928	0.993	0.986
EthanolConcentration	0.772	0.260	0.3954	0.369	0.734
FaceDetection	0.660	0.557	0.5074	0.589	0.627
FingerMovements	0.530	0.501	0.590	0.560	0.520
HandMovementDirection	0.473	0.245	0.3378	0.284	0.595
Handwriting	0.548	0.274	0.2694	0.520	0.356
Heartbeat	0.732	0.768	0.7268	0.741	0.780
LSST	0.643	0.673	0.3362	0.603	0.573
Libras	0.933	0.847	0.8556	0.844	0.911
MotorImagery	0.540	0.503	0.500	0.460	0.500
NATOPS	0.894	0.835	0.8778	0.917	0.856
PEMS-SF	1.000	0.999	0.3642	0.908	1.000
PenDigits	0.979	0.952	0.6209	0.920	0.967
PhonemeSpectra	0.290	0.284	0.1786	0.251	0.265
RacketSports	0.908	0.806	0.8487	0.908	0.882
SelfRegulationSCP1	0.891	0.820	0.8567	0.758	0.860
SelfRegulationSCP2	0.500	0.418	0.4778	0.472	0.500
StandWalkJump	0.467	0.363	0.400	0.400	0.400
UWaveGestureLibrary	0.928	0.775	0.875	0.944	0.925
Average accuracy	0.748	0.667	0.607	0.690	0.726
Wins	19	1	4	5	6

Table 9: Summary of machine-learning MTSC algorithm evaluation.

Year	Algorithm	Reference	UEA datasets	Compared against	Source code
2020	CIF	Middlehurst et al. (2020)	26	6	Yes ^a
2020	MSAX	Anacleto et al. (2020)	14	3	No
2020	MI-FSS	Ircio et al. (2020)	17	2	No
2020	MSTS	Kathirgamanathan and Cunningham (2020)	6	2	No
2021	CMFMTS	Baldán and Benítez (2021)	30	11	Yes ^b
2021	F4	López-Oriona and Vilar (2021)	6	4	No
2021	HC2	Middlehurst et al. (2021)	26	4	Yes ^c
2021	MTSPF	Zhang et al. (2021a)	20	9	No
2022	RT	Bahri et al. (2022)	30	4	Yes ^d
2022	PETSC	Feremans et al. (2022)	26	11	Yes ^e
2024	EP/IEP	Campagner et al. (2024)	24	8	Yes ^f
2024	FS-MVC	He et al. (2024)	26	6	No
2024	Z-Time	Lee et al. (2024)	20	12	Yes ^g
2024	EE-TSS	Patharkar et al. (2024)	6	2	No
2024	BORF	Spinnato et al. (2024)	30	20	Yes ^h
2024	BOP	Wang et al. (2024d)	18	7	No
2025	FedDict	Liang et al. (2025a)	20	6	No
2025	GSM	Panwar (2025)	24	1	No
2026	TimeCIEL	Levy et al. (2026)	6	3	No

^a <https://timeseriesclassification.com/CIF.php>

^b <https://github.com/fjbaldan/CMFMTS>

^c <https://www.timeseriesclassification.com/HC2.php>

^d <https://github.com/omarbahri/RuleTransform>

^e https://bitbucket.org/len_feremans/petsc

^f <https://github.com/AndreaCampagner/scikit-cautious>

^g <https://github.com/zedshape/ztime>

^h <https://github.com/fspinna/borf>

Table 10: Accuracy of six convolutional MTSC classifiers on the UEA archive datasets used to generate Figure 6. Note that HDTCNet, TPS and FreConvNet source code contains leakage from test to train.

Dataset	Disjoint-CNN	XCM	TPS	FreConvNet	HDTCNet	LITEMVTime
ArticularyWordRecognition	0.984	0.983	0.990	0.993	0.993	0.973
AtrialFibrillation	0.375	0.467	0.467	0.533	0.467	0.133
BasicMotions	1.000	1.000	1.000	1.000	1.000	1.000
CharacterTrajectories		0.995	0.999			0.996
Cricket	0.999	1.000	1.000	1.000	1.000	0.986
DuckDuckGeese	0.644	0.700	0.740			0.180
EigenWorms	0.593	0.435	0.954			0.939
Epilepsy	0.989	0.993	0.978	0.991	1.000	0.993
ERing	0.964	0.133	0.948		0.926	0.844
EthanolConcentration	0.309	0.346	0.361	0.360	0.326	0.692
FaceDetection	0.567	0.639	0.648	0.651	0.667	0.610
FingerMovements	0.604	0.600	0.630	0.638	0.670	0.560
HandMovementDirection	0.324	0.446	0.446	0.541	0.486	0.351
Handwriting	0.321	0.412	0.587	0.575	0.683	0.400
Heartbeat	0.759	0.776	0.780	0.776	0.803	0.615
InsectWingbeat		0.105	0.695	0.766		0.617
JapaneseVowels		0.986	0.992	0.986	0.992	0.965
Libras	0.916	0.844	0.900	0.959	0.961	0.894
LSST	0.547	0.612	0.652	0.699	0.438	0.664
MotorImagery	0.569	0.540	0.650	0.643		0.530
NATOPS	0.967	0.978	0.956	0.984	0.989	0.900
PenDigits	0.995	0.991	0.991	0.988	0.993	0.989
PEMS-SF	0.817	0.757	0.815		0.896	0.792
PhonemeSpectra	0.282	0.225	0.326	0.261	0.355	0.158
RacketSports	0.846	0.895	0.921	0.904	0.928	0.737
SelfRegulationSCP1	0.903	0.878	0.877	0.928	0.928	0.730
SelfRegulationSCP2	0.538	0.544	0.594	0.594	0.596	0.550
SpokenArabicDigits		0.995	0.998	0.997	0.999	0.986
StandWalkJump	0.436	0.400	0.667	0.433	0.533	0.667
UWaveGestureLibrary	0.924	0.894	0.913	0.932	0.923	0.847
Average accuracy	0.699	0.686	0.782	0.765	0.782	0.710
Wins	3	2	8	9	17	2

Table 11: Summary of convolutional MTSC papers and reported source-code availability.

Year	Algorithm	Reference	UEA datasets	Compared against	Source code
2020	CA-SFCN	Hao and Cao (2020)	1	5	Yes ^a
2020	CT_CAM	Cheng et al. (2020)	15	9	No
2020	TapNet	Zhang et al. (2020)	15	8	Yes ^b
2021	Disjoint-CNN	Foumani et al. (2021)	26	9	Yes ^c
2021	GM-FCN	Tai et al. (2021)	1	9	No
2021	XCM	Fauvel et al. (2021)	30	10	Yes ^d
2022	AC-LSTM	Zhang et al. (2022)	11	5	No
2022	DA-Net	Chen et al. (2022)	14	11	Yes ^e
2022	dCAM	Boniol et al. (2022)	17	10	Yes ^f
2022	DSN	Xiao et al. (2022)	0	3	Yes ^g
2022	Multichannel CNN	Oh (2022)	5	5	No
2022	TCRAN	Zhu et al. (2022)	15	10	No
2023	Adacket	Zhang et al. (2023)	3	2	Yes ^h
2023	Conv-GLU	Liu et al. (2023)	23	4	No
2023	POPNASv3	Falanti et al. (2023)	2	3	Yes ⁱ
2024	AdaFSNet	Wang et al. (2024b)	21	4	No
2024	TPS	Abbasi and Saeedi (2024)	30	12	Yes ^j
2025	DCA	Haboub et al. (2025)	5	8	No
2025	DM-FCN	Zhang et al. (2025)	30	6	Yes ^k
2025	FQN	Orka et al. (2025)	6	8	Yes ^l
2025	FreConvNet	Tang et al. (2025b)	25	13	Yes ^m
2025	HDTCNet	Gu (2025a)	25	16	Yes ⁿ
2025	LITEMVTime	Ismail-Fawaz et al. (2025)	30	6	Yes ^o
2025	RE-ModernTCN	Akacik and Hoogendoorn (2025)	1	10	Yes ^p
2025	SCNet	Wu et al. (2025b)	10	7	No
2025	TEAM	Qiu et al. (2025b)	15	9	No

^a https://github.com/huipingcao/nmsu_yhao_ijcai2020

^b <https://github.com/kdd2019-tapnet/tapnet>

^c <https://github.com/NavidFoumani/Disjoint-CNN>

^d <https://github.com/XAISeries/XCM>

^e <https://github.com/Sample-design-alt/DANet>

^f <https://helios2.mi.parisdescartes.fr/~themisp/dCAM/>

^g <https://github.com/QiaoXiao7282/DSN>

^h <https://github.com/jrzhang33/Adacket>

ⁱ <https://zenodo.org/record/7439022>

^j <https://github.com/mehryar72/TimeSeriesClassification-TPS>

^k <https://github.com/wz16/DMFCN>

^l <https://doi.org/10.5281/zenodo.15486109>

^m <https://github.com/yingxiatang/FreConvNet>

ⁿ <https://github.com/Guuuli/HDTCNet>

^o <https://github.com/MSD-IRIMAS/LITE>

^p <https://github.com/onderakacik/RE-ModernTCN>

Table 12: Summary of graph-based MTSC papers and reported source-code availability.

Year	Algorithm	Reference	UEA datasets	Compared against	Source code
2022	MTPool	Duan et al. (2022)	10	10	Yes ^h
2022	SimTSC	Zha et al. (2022)	4	6	Yes ^a
2024	CATodyNet	Gui et al. (2024)	22	13	Yes ^b
2024	HyperTime	Younis and Ahmadi (2024)	26	13	Yes ^c
2024	MagNet	Hong et al. (2024)	30	13	No
2024	TF-Net	Lei et al. (2024)	24	10	No
2024	RTNet	Gu (2024)	15	7	No
2025	CTSA	Du and Chen (2025)	17	10	Yes ^e
2025	DET-DyAGCN	Gui (2025)	30	15	Yes ^f
2025	DVG-CL	Xiao et al. (2025b)	11	9	Yes ^g
2025	GSTT	Zhang et al. (2026)	30	12	No
2025	KDP-MHL	Ma et al. (2025a)	24	11	No
2025	MH-TFFN	Jiang et al. (2025)	4	11	No
2025	MuPaST	Li et al. (2025)	26	10	No
2025	RP-GCN	Kang et al. (2025)	23	10	No
2025	STOP	Liu et al. (2025)	29	8	No
2025	TV-Net	Yue et al. (2025)	30	12	No

^a <https://github.com/daochenzha/SimTSC>

^b <https://github.com/HQYWY/CATodyNet>

^c <https://gitfront.io/r/user-9180183/WXPgc7aQjSKE/HyperTime/>

^e https://github.com/xiaodul999/HGRL-master_root

^f <https://github.com/ywyhq/DET-DyAGCN>

^g <https://github.com/ASL-XZY/DVG-CL>

^h <https://github.com/RRRussell/MTPool>

Table 13: Summary of transformer MTSC papers and reported source-code availability.

Year	Algorithm	Reference	UEA datasets	Compared against	Source code
2022	FlowFormer	Wu et al. (2022)	10	13	Yes ^a
2022	FourierFormer	Nguyen et al. (2022b)	10	1	Yes ^b
2022	GFISHformer	Nguyen et al. (2022a)	15	1	Yes ^c
2023	FormerTime	Cheng et al. (2023)	10	11	Yes ^d
2023	MCTNet	Wu et al. (2023b)	26	9	No
2023	MF-Net	Du et al. (2023)	14	12	Yes ^e
2023	Primal-Attention	Chen et al. (2023)	10	10	Yes ^f
2024	AutoLDT	Wang et al. (2024c)	4	7	No
2024	ConvTran	Foumani et al. (2024b)	30	5	Yes ^g
2024	CTNet	Zhang et al. (2024)	30	9	Yes ^h
2024	DCSformer	Wang et al. (2024a)	17	8	No
2024	DTCM	Xiao et al. (2024b)	29	20	No
2024	MEDT	Li et al. (2024)	5	5	No
2024	MSTformer	Wu et al. (2024)	30	5	No
2024	STIFformer	Zhang and Li (2024)	30	8	No
2024	ShapeFormer	Le et al. (2024)	30	12	Yes ⁱ
2025	CAB	Nguyen et al. (2025)	10	14	No
2025	CSTFormer	Qiu et al. (2025a)	30	5	No
2025	DTWformer	Yu et al. (2025)	16	6	No
2025	Hformer	Tang et al. (2025a)	30	9	No
2025	KATN	Xiao et al. (2025a)	30	6	No
2025	MD-Former	Du et al. (2025)	30	11	No
2025	PBC-CSA	Gu (2025b)	17	11	No
2025	PatchMTSC	Wei et al. (2025)	30	19	Yes ^j
2025	SGN	Ying et al. (2025)	25	8	Yes ^k
2025	Sliding-window Trans- former + DTW similar- ity	Wu et al. (2025a)	16	2	No
2026	TFTN	Huang et al. (2026)	19	8	No

^a <https://github.com/thuml/Flowformer>^b https://github.com/minhtannguyen/FourierFormer_NeurIPS^c <https://github.com/minhtannguyen/FishFormer>^d <https://github.com/mingyue-cheng/formertime>^e <https://github.com/dumingsen/MF-Net>^f <https://github.com/yingyichen-cyy/PrimalAttention>^g <https://github.com/Navidfoumani/ConvTran>^h <https://github.com/zhangda1018/CTNet>ⁱ <https://github.com/xuanmay2701/shapeformer>^j <https://github.com/YanxuanWei/PatchMTSC>^k <https://github.com/colison/SGN>

Table 14: Summary of foundation-style MTSC papers and reported source-code availability.

Year	Algorithm	Reference	UEA datasets	Compared against	Source code
2021	TST	Zerveas et al. (2021)	11	5	Yes ^a
2022	TARNet	Chowdhury et al. (2022)	30	14	Yes ^b
2022	TS2Vec	Yue et al. (2022)	29	5	Yes ^c
2023	FEAT	Kim et al. (2023)	30	6	No
2023	TimesNet	Wu et al. (2023a)	10	15 ^q	Yes ^d
2024	HSC-MAE	Yu et al. (2024)	9	7	No
2024	LPTM	Kamarthi and Prakash (2024)	30	9	Yes ^e
2024	LSTST	Jang et al. (2024)	10	19	Yes ^f
2024	ModernTCN	Luo and Wang (2024)	10	14	Yes ^g
2024	TimesURL	Liu and Chen (2024)	30	7	Yes ^h
2024	Series2Vec	Foumani et al. (2024a)	30	5	Yes ⁱ
2025	CBD	Han et al. (2025)	10 ^r	9	No
2025	CauLLM	Wu et al. (2025c)	4	9	Yes ^j
2025	LETS-C	Kaur et al. (2025)	30 ^s	27	No
2025	LLMFew	Chen et al. (2025b)	10	11	Yes ^k
2025	PETS	Ma et al. (2025b)	10	16	No
2025	S2TS-LLM	Qin et al. (2025)	10	9	Yes ^l
2025	TableTime	Wang et al. (2025)	10	17	Yes ^m
2025	TFAS	Tirrat and Lee (2025)	10	17	Yes ⁿ
2025	TSFM Adapters	Ilbert et al. (2025)	12	8 ^t	Yes ^o
2026	HiTime	Tao et al. (2026)	10	14	Yes ^p

^a https://github.com/gzerveas/mvts_transformer

^b <https://github.com/ranakroychowdhury/TARNet>

^c <https://github.com/yuezhihan/ts2vec>

^d <https://github.com/thuml/TimesNet>

^e <https://github.com/AdityaLab/LPTM>

^f <https://github.com/HyeongWon-Jang/LSTST>

^g <https://github.com/luodhhh/ModernTCN>

^h <https://github.com/Alrash/TimesURL>

ⁱ <https://github.com/Navidfoumani/Series2Vec>

^j <https://github.com/ShuangWu111/CauLLM>

^k <https://github.com/junekchen/llm-fewshot-mtsc>

^l <https://github.com/JianyangQin/S2TS-LLM>

^m <https://github.com/ForestsKing/TableTime>

ⁿ <https://github.com/thuml/Time-Series-Library>

^o <https://github.com/abenechehab/AdaPTS>

^p <https://github.com/Xiaoyu-Tao/HiTime>

^q Uses a shared cross-task baseline suite, so the comparator count is not MTSC-only.

^r The yearly summary does not state the UEA-only subset explicitly.

^s The summary states 33 UEA datasets, but only 30 are named, so the table reports 30.

^t The comparator count includes adapter and head variants, not only distinct external baselines.

Table 15: Summary of other deep MTSC papers and reported source-code availability.

Year	Algorithm	Reference	UEA datasets	Compared against	Source code
2020	STSP-SNN	Fang et al. (2020)	7	4	No
2021	ShapeNet	Li et al. (2021)	30	7	Yes ^a
2021	GMSM _{red}	Medico et al. (2021)	10	5	No
2021	RTFN	Xiao et al. (2021)	30	15	No
2021	mGRN	Zhang et al. (2021b)	6	9	Yes ^b
2021	SMATE	Zuo et al. (2021)	30	10	Yes ^c
2023	WHEN	Wang et al. (2023)	30	8	No
2023	GP-HLS	Ye and Ma (2023a)	6	3	No
2023	LBP4MTS	Ye and Ma (2023b)	10	4	No
2023	DSC	Zhao et al. (2023)	3	2	No
2024	Chimera	Behrouz et al. (2024)	10	17	No
2024	AGDM	Ding et al. (2024)	11	9	No
2024	FDNet	Gao et al. (2024)	23	10	No
2024	MTSFNet	Huang et al. (2024)	10	10	No
2024	DECM (ODE-RGRU)	Jeong et al. (2024)	30	10	Yes ^d
2024	DSDCLNet	Liu et al. (2024)	30	13	No
2024	RankSCL	Ren et al. (2024)	30	8	Yes ^e
2024	VQShape	Wen et al. (2024)	29	13	Yes ^f
2024	DKN	Xiao et al. (2024a)	30	18	No
2024	TSec	Yao et al. (2024)	26	19	Yes ^g
2025	TSCMamba	Ahamed and Cheng (2025)	30	N/A	No
2025	LETO	Behrouz et al. (2025)	10	N/A	No
2025	SFC	Chen et al. (2025a)	12	9	No
2025	DAPNet	Gao (2025)	10	10	No
2025	TCFNet	Liang et al. (2025b)	26	9	No
2025	RFACL	Lin et al. (2025)	10	10	Yes ^h
2025	MPTSNet	Mu et al. (2025)	25	21	Yes ⁱ
2025	GrowingNN	Swiderski et al. (2025)	N/A	N/A	Yes ^j
2025	FreRA	Tian et al. (2025)	30	10	Yes ^k
2025	NI-MTSC	Yang et al. (2025)	18	7	No
2025	RTF	Yuan et al. (2026)	30	15	Yes ^l
2026	MatrixMixer	Azizi and BabaAli (2026)	29	N/A	Yes ^m
2026	WAVESTATE	Li et al. (2026)	20	17	No

N/A indicates that the value was not clearly stated in the extracted paper text.

^a Paper link is dead: <http://alturl.com/d26bo>; project files remain available at <https://www.comp.hkbu.edu.hk/~csgzli/mtsc/>

^b <https://github.com/yaquanzhang/mGRN>

^c <https://github.com/JingweiZuo/SMATE>

^d <https://github.com/ku-milab/Deep-Efficient-Continuous-Manifold-Learning>

^e <https://github.com/UConn-DSIS/Rank-Supervised-Contrastive-Learning-for-Time-Series-Classification>

^f <https://github.com/YunshiWen/VQShape>

^g <https://github.com/ZJU-DAILY/TSec>

^h <https://github.com/LinHuai01/RFACL>

ⁱ <https://github.com/MUYang99/MPTSNet>

^j <https://pypi.org/project/growingnn/>

^k <https://github.com/lukewys/FreRA>

^l <https://anonymous.4open.science/r/RTF->

^m <https://github.com/mhmdmdiaz-izi/matrix-mixer-tsc>

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